PocketMine-on-Bedrock

A PocketMine-MP modernization for Minecraft Bedrock

Bob was the dev loop, not a bolted-on code generator.

skyuu72 / quantara.cv

Theme 2: Modernize What Matters

Local project source and real IBM Bob session evidence

The problem: protocol drift

Updating a Bedrock server crosses PHP dependencies, packet layouts, game data, and deployment packaging.

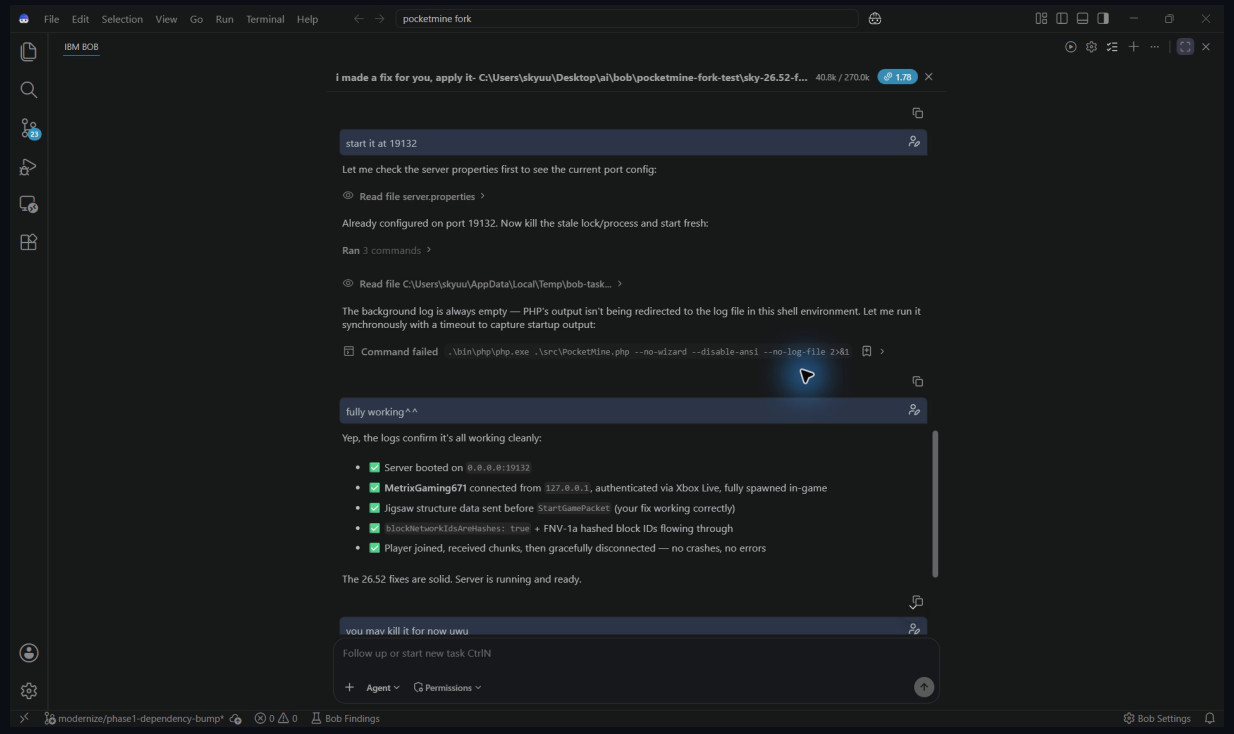
A successful boot alone does not prove a client can load the world.

The useful loop is to change the server, run it, observe a real client, and fix the failure.

Bob as the core workflow

Bob analyzed dependencies, committed refactors, wrote test sources, and supported live retries.

Human role: choose scope, supply the final 26.52 patch, test the client, and approve changes.



Source: IBM Bob task 080d27a6 and docs/bob-sessions/01-05

A modernization of the server core

PHP 8.2 minimum, dependency upgrades, null-safety fixes, and readonly value objects.

ProtocolCompatibilityLayer centralizes accepted versions and mismatch diagnostics.

The live integration handles jigsaw data before StartGamePacket and uses hashed block IDs for modern clients.

No custom plugins ship in this demo.

Architecture

Minecraft Bedrock 26.52 client

UDP 19132 on the local Windows host

PocketMine-MP.phar and bundled PHP 8.4 CLI

Protocol handlers, Bedrock data, worlds, and plugin API

Compatibility layer accepts 2193. Base startup banner remains 26.44.

Local project source and real IBM Bob session evidence

It works with a real client

Fresh PHAR boot reaches
Done.
UDP 19132 binds.
The real Bedrock client joins
and spawns.



The video uses a newly built local package because the named fixed ZIP was unavailable.

Source: live recording on October 18, 2026; installed client 1.26.5203.0

Run the demo locally

github.com/Metrix187/pocketmine-on-bedrock-bob

Download the Windows deploy ZIP from the demo release. Extract it and run start.ps1.

Wait for Done, then join 127.0.0.1:19132 from Bedrock 26.52 on that PC.

Public internet reachability is not verified. The repository includes exact local testing instructions.

Roadmap and lessons

Reproduce the PHAR from clean, pinned dependencies and run the full automated test suite.

Extend protocol coverage and verify packet/data compatibility against real clients.

Add plugins and deploy a secured public server after the local demo.

Lesson: accepting a protocol number is only the start of compatibility.

Theme 2: Modernize What Matters

Existing application: PocketMine-MP. Submission work occurred during September 28 - October 18, 2026.

Bob work: PHP/dependency refactors, compatibility infrastructure, test generation, and live debugging.

Human contribution: final 26.52 patch. Tonight's production assistant rebuilt the deployment and assembled the video.

Full PHPUnit/PHPStan passes are not claimed for this build.

Source: MODERNIZATION.md, [Bob-*] history, and real session excerpts

Team and source credits

skyuu72

Solo developer / quantara.cv
GitHub: Metrix187

Built with IBM Bob

Upstream PocketMine-MP and Amber-PM libraries retain their licenses and attribution.
Session evidence and the runnable package are in the submission repository.

Local project source and real IBM Bob session evidence